

Subject Description Form

Subject Code	EE4024 / EE4011A / EE4011B
Subject Title	Industrial Computer Applications
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Introduce the applications of advanced computing techniques in solving industrial problems. The topics include: embedded computing system; Bidirectional Digital Power System; Internet of Things (IoT) applications and 5G/6G Digital Communication Infrastructure.
Subject Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Apply advanced computing techniques to address industrial challenges - Understand the significance of computing systems in industrial applications. - Develop logical thinking skills, analyze data effectively, and present results clearly in writing.
Subject Synopsis/ Indicative Syllabus	Embedded Computer systems: Introduction to embedded systems, embedded system design, . These systems are important for functionality, control and monitoring of various devices and processes in diverse industries. Bidirectional Digital Power System: This system leverages a high performance from computing control devices, rendering a high-power bidirectional power conversion through AC/DC conversion techniques. The high-efficient system provides distinct power management, ensuring industries such as renewable energy, electric vehicle (EV), industrial automation, can opt for the optimal configuration to meet their unique application needs. 5G/6G Digital Communication Infrastructure: These advanced wireless communication technology focuses on enhanced mobile broadband, ultra-reliable low latency and massive machine-type communication. Applying the Air interface technology, the frequency of radio waves to use for transmission, what kind of antenna to use, how to encode information into the radio waves, advanced modulation, multiplexing technologies, and so on can be specified in the 5G digital communication. IoT and Mobile applications: IoT design and implementation. Introduction to server-side and client-side applications and the MQTT platform. Mini-project: Apply one of the above computing topics to solve an engineering problem.
Teaching/Learning Methodology	Lectures and tutorials serve as the main methods for teaching fundamental concepts and theories. Experiences in design and practical applications are given through the execution of a mini-project, where the students are expected to solve design problems with real-world constraints and attain pragmatic solutions with critical and analytical thinking.

	Teaching/Learning Methodology		Outcomes		
			a	b	c
	Lectures	✓	✓		
	Tutorials	✓	✓		
	Mini-project	✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Examination	60%	✓	✓	✓
	2. In-class Test	15%	✓	✓	✓
	3. Mini-project	18%	✓	✓	✓
	4. Exercise	7%	✓	✓	
	Total	100%			
	The assessment includes a written examination at the end of the semester, a mid-semester test, and a mini-project focused on an industrial computing application. The mini-project requires a study report that investigates the feasibility of the application, explains any failures, and identifies opportunities for future enhancements and improvements.				
Student Study Effort Expected	Class contact:				
	▪ Lecture/Tutorial		39 Hrs.		
	▪ Mini-project presentation		3 Hrs.		
	Other student study effort:				
	▪ Mini-project implementation and report writing		23 Hrs.		
	▪ Self-study		40 Hrs.		
	Total student study effort		105 Hrs.		
Reading List and References	Reference books and online materials:				
	1. T. Cox, et al., Getting Started with Python for the Internet of Things, Maker Media, Inc, 2019.				
	2. U. Meyer-Baese, Embedded microprocessor system design using FPGAs, Springer, 2021.				
	3. E. White, Making Embedded Systems: Design Patterns for Great Software, O’Reilly, 2011.				
	4. M. Beyeler, Machine Learning for OpencvCV: Intelligent image processing with Python, Packt Publishing, 2017.				
	5. Y. L. Prasad, Big Data Analytics Made Easy, Notion Press, 2016				
	6. T. White, Hadoop: The Definitive Guide, 3 rd Ed, O’Reilly, 2012				